

ABSTRACT

OBJECT DETECTION FROM AN IMAGE:

Object detection is one of the most important applications of computer vision and deep learning. It is used to identify and locate objects present in an image automatically. The main objective of this project is to detect various objects from input images with improved accuracy and efficiency using deep learning techniques.

In this project, image processing and object detection algorithms are implemented using Python along with libraries such as OpenCV, TensorFlow, and NumPy. The system takes an input image, processes it, and identifies different objects by drawing bounding boxes and displaying labels around the detected objects. The project is trained using pre-trained deep learning models that help in improving detection accuracy and reducing processing time.

Object detection has wide applications in various fields such as surveillance systems, autonomous vehicles, medical imaging, security monitoring, and smart traffic management. This project demonstrates how artificial intelligence and deep learning can be effectively used for real-world image analysis and automation tasks.

The developed system successfully detects objects from images and provides accurate results, making it useful for real-time computer vision applications. Future improvements can include real-time video detection, enhanced accuracy, and deployment as a web or mobile application.